

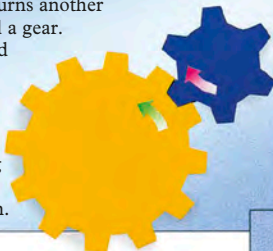


Easy



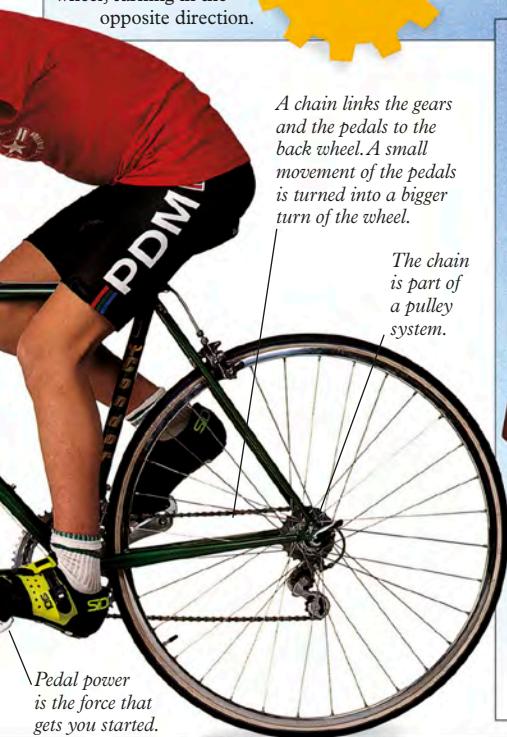
Get into gear

A toothed wheel that turns another toothed wheel is called a gear. Gears change the speed or direction of the moving part of a machine. The small wheel spins around twice as fast as the big wheel, turning in the opposite direction.



A chain links the gears and the pedals to the back wheel. A small movement of the pedals is turned into a bigger turn of the wheel.

The chain is part of a pulley system.



Pedal power is the force that gets you started.

Uphill climb

Slopes are simple machines. They make it easier to move things onto a higher or lower level. A screw is a kind of slope. If you could unwind the spiral grooves on a screw, they would flatten out as a slope.



Light as a feather?

Lifting up a weight with your bare hands is hard work. It is much easier to lift a weight by pulling down on the ropes of a pulley. If you double the number of pulley wheels, the same amount of effort will lift twice as many weights.

